

# A perturbation of the seven-eighths zero-free half-plane

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## Abstract

We examine the numerical endpoint in the compensated-probe proof of the seven-eighths zero-free half-plane in [1]. The reflected-row estimate has room to increase the total prime length, while the high-side endpoint certificate has a fixed positive margin. Keeping the same probe and changing its lengths gives, from the analytic estimates of [1], a zero-free half-plane with boundary  $174999/200000 = 0.874995$ . This note records the parameter changes and the extensions of the estimates used at the boundary.

## 1 Statement and input

Let  $F = \mathbb{Q}(\sqrt{-3})$ . As in [1, Equation (1.1b)], let  $\beta_*$  be the supremum of  $1/2$  and the real parts of zeros of primitive finite-order Hecke  $L$ -functions over  $F$  in the critical strip. We use the seven-eighths theorem of [1], so  $\beta_* \leq 7/8$ . We also use the arithmetic identities, detector estimates, reflected-energy lemma, additive Gram bound, and continuation criterion proved there. Section 2 checks the changes needed when its Part II geometry is perturbed. Section 3 checks the high comparison and the analytic domains.

**Theorem 1.** *Every finite-order Hecke  $L$ -function over  $F$  and every Dirichlet  $L$ -function, including the Riemann zeta function, has no zero in*

$$\operatorname{Re} s > B := \frac{174999}{200000} = \frac{7}{8} - \frac{1}{200000}.$$

*The usual pole at  $s = 1$  for a principal character is allowed.*

Put

$$\begin{aligned} t &= \frac{3}{100000}, & \ell &= \frac{1}{6} + t, & b &= \frac{1}{8}, \\ l_x &= \frac{17}{48} - \frac{t}{2}, & l_y &= \frac{23}{48} - \frac{t}{2}, & h &= \frac{13}{16} + \frac{3t}{2}, \\ M &= l_x + l_y = \frac{5}{6} - t. \end{aligned} \tag{1}$$

These lengths satisfy  $l_y - l_x = b$ ,  $M + \ell = 1$ , and  $h = 1 - l_x + \ell$ . The general Mellin-exponent identity in [1, Part II, compensated low estimate] therefore remains

$$C(s) = \frac{l_x}{2} + s - 1 + \frac{h}{6} = s - \frac{11}{16}. \tag{2}$$

Choose a fixed even number  $K$  of disjoint prime slots, each of length  $\ell/K$ , as in [1, Section 5, “Order of choices”]. The slot count may be enlarged to meet the same mesh requirements. The finite physical probe and its exact Poisson identity in [1] make sense with the lengths in (1). All its arithmetic factors are unchanged.

## 2 The low comparison

**Lemma 2.** *The normalized compensated probe of [1], formed with (1), satisfies for every  $\varepsilon > 0$*

$$|J_\eta(Z)| \ll_{\eta,\varepsilon} Z^{3/16-t/4+\varepsilon}.$$

*Proof.* Fix a rescaled subset of slots of total logarithmic length  $0 \leq d \leq \ell$ . In the notation of the compensated low estimate of [1], its row and surviving-slot lengths are

$$M' = M - 2d, \quad \ell' = \ell - d.$$

The proof of the reflected-row norm in [1, Part II, compensated low estimate] first bounds one branch by  $M'$  and the other by  $M' + (1 + 3\ell' - 2M')/4$ , up to arbitrary small losses. Its general reflected-energy lemma permits these fixed bounded lengths. Substitution gives the squared row norm

$$\sum_{0 < N \leq m \leq Z^{M'}} |B_{m,\sigma}^J(Z)|^2 \ll Z^{M' + ((d-1/6+5t)/4)_+ + \varepsilon}. \quad (3)$$

The additive Gram estimate is available throughout this range. Its scale conditions follow from

$$\begin{aligned} l_x - \ell &= \frac{3}{16} - \frac{3t}{2} > 0, & l_y - \ell &= \frac{5}{16} - \frac{3t}{2} > 0, \\ M - 2\ell &= \frac{1}{2} - 3t > 0, & l_y - \ell - \frac{11b}{6} &= \frac{1}{12} - \frac{3t}{2} > 0. \end{aligned}$$

Thus its  $P_a^2/Y'$  term is absorbed by  $P_a^{1/6}$ , exactly as in [1, Part II, additive Gram bound]. Cauchy–Schwarz, the count of rescaled tuples, and their coefficients add to the basic low exponent

$$-d + \left( \frac{d - 1/6 + 5t}{8} \right)_+.$$

This is nonpositive for  $0 \leq d \leq \ell$ . Indeed, its positive-part argument is negative until  $d > 1/6 - 5t$ , and thereafter  $d - 1/6 + 5t < 8d$ . The empty subset has exponent zero. The direct bound is consequently  $Z^{l_x/2+b/12+\varepsilon}$ .

The prime normalizer in [1, Part II, principal normalizer] is nonzero and has inverse  $Z^{o(1)}$  for every fixed positive slot length. Finally,

$$\frac{l_x}{2} + \frac{b}{12} = \frac{3}{16} - \frac{t}{4}.$$

This proves the normalized estimate. □

The signal at the proposed boundary is

$$C(B) = \frac{3}{16} - \frac{1}{200000}, \quad C(B) - \left( \frac{3}{16} - \frac{t}{4} \right) = \frac{1}{400000} > 0. \quad (4)$$

## 3 The high comparison

We first record the extension of the Euler regions used by the contour arguments. The local factors below are those of [1, Equations (5.13a)–(5.13c)].

**Lemma 3.** *The full compensated correction is holomorphic, with the bounds needed for principal-row contour moves, on the second Euler region of [1] with its condition  $\operatorname{Re} s \geq 7/8$  replaced by  $\operatorname{Re} s \geq B$ . The principal local quotient there satisfies*

$$G_p/H_p = -1 + O((Np)^{-B}).$$

*The small-row contour can use the first Euler region with  $\epsilon_0 = 37/100$ .*

*Proof.* The exact local defect formula in [1, local Euler lemma] bounds  $H_p - 1$  in the original second region by  $O(Q^{-363/200})$  for  $p \nmid u$  and  $O(Q^{-33/40})$  for  $p \mid u$ , where  $Q = Np$ . Every local exponent involving  $\operatorname{Re} s$  has coefficient of absolute value at most six. Moving its lower boundary left by  $1/200000$  therefore gives the respective bounds

$$O(Q^{-363/200+3/100000}) \quad \text{and} \quad O(Q^{-33/40+3/100000}).$$

The first exponent is less than  $-1$ . The unramified product converges normally, while the ramified product has the same  $O_\varepsilon((Nu)^\varepsilon)$  bound. For  $u = 1$  its tail still tends to zero faster than  $P_0^{-4/5}$ . Enlarging the fixed excluded set therefore gives the same uniform nonvanishing of the principal product.

The four errors for the principal selected factor have exponents  $-\operatorname{Re} s$ ,  $-6 \operatorname{Re} z$ ,  $4 - 5 \operatorname{Re} s - 6 \operatorname{Re} z$ , and  $1 - \operatorname{Re} w - 6 \operatorname{Re} z$ . On the extended region they are all at most  $-B$ . The formula for  $G_p$  has no division by  $P_p$  or  $H_p$ , so the full finite selected-tuple correction remains holomorphic at their zeros. The bounds above and finite slot count give the all-height majorant used in the principal contour proof of [1].

For small rows,  $\beta_* > B$  implies  $\beta_* + e + 1/2 > 1 + 37/100$ . The first Euler region of [1] allows any positive  $\epsilon_0$ , so its unchanged local estimates apply to the line  $(\beta_* + e, 1/2, 17/50)$ . The central bin contours already lie in its region with  $\epsilon_0 = 10e$ . These are the only analytic domain uses of the original  $7/8$  lower bound in the shared contour lemmas. Repeating their proofs with the extended region gives the same principal, bin, and outer-row contour estimates.  $\square$

**Lemma 4.** *If  $B < \beta_* \leq 7/8$ , then for some  $\sigma > 0$  independent of the target character,*

$$|J_\eta(Z) - f_\eta(Z)| \ll_\eta Z^{C(\beta_*)-\sigma},$$

*where  $f_\eta$  is the principal Mellin signal from [1] with (2).*

*Proof.* The fourth-moment lemma of [1] permits  $\kappa = 3/4$ , because its zero-free input is  $\beta_* \leq (1 + \kappa)/2 = 7/8$ . Every detected bin has  $\delta = 2a - 1 \leq 2\beta_* - 1 \leq 3/4$ . Its plain prime capacity is thus the baseline capacity in the proof of [1, Part II, capacity comparison], with no positive  $\Delta/4$  surcharge. The row exponent is bounded by  $R_*(\delta, x)$  plus an arbitrarily small loss, with  $0 \leq x \leq 1/2$  and  $q = x\delta$ . The same selected-prime argument applies since  $\ell/h > 8/39 > 7/37$  and  $5\ell - h > 1/48$ . A fixed sufficiently fine even slot mesh meets all rounding conditions.

At row length  $d = h$ , the general high exponent relative to  $C(B)$  is

$$E_B(h) = a - B + h \left( \frac{17}{50} - \frac{1}{6} \right) - a\ell_y - \frac{\ell}{2} + q\ell + h \left( R + \frac{\delta}{2} - \frac{17}{50} \right). \quad (5)$$

The endpoint certificate of [1, Part II, compensated endpoint certificate], evaluated at its original geometry, bounds this expression with  $B = 7/8$  and  $R = R_*$  by  $-49/440640$  for the larger range  $\delta \leq 5/6$ . Holding  $a, \delta, q, R$  fixed while changing (1) gives

$$E_B(h) - E_{7/8}(13/16) = \frac{1}{200000} + t \left( -\frac{1}{2} + \delta + q + \frac{3R}{2} \right). \quad (6)$$

The source bounds  $R_* \leq 17/12$ . Since  $\delta \leq 3/4$  and  $q \leq \delta/2$ , the right side of (6) is at most

$$\frac{1}{200000} + \frac{11t}{4} = \frac{7}{80000}.$$

Thus the central endpoint retains the explicit margin

$$-E_B(h) \geq \frac{49}{440640} - \frac{7}{80000} > 0. \quad (7)$$

The frequency slope is positive, so this endpoint bounds all moderate rows with  $d \leq h$ . Extending to  $h + \zeta$  costs at most  $2\zeta$ , and  $\zeta$  can be chosen below the margin in (7).

The floor and intermediate-row margins in [1] are respectively  $7/1200$  and  $49/14400$  at the original geometry. Their changes are bounded by the same  $7/80000$  and are therefore harmless. The small-row margin is  $63/800$ . Its change is at most  $51t/100$ , since  $h(17/50 - 1/6) - l_y/2$  has derivative  $51/100$  with respect to  $t$ . For large rows, the absolute contour can still be taken sufficiently far right after choosing  $\zeta$ . Lemma 3 supplies the principal contour, whose margins  $l_y/20$  and  $h/600$  remain positive. Its local approximation supplies a positive power saving in each fixed slot. The prime normalizer costs only  $Z^{o(1)}$ .

All detector, mesh, real-contour, and normalizer losses can be chosen below these fixed positive margins before the target character is fixed. The late height choice in [1] then gives one  $\sigma > 0$  for every target, as claimed.  $\square$

*Proof of Theorem 1.* Suppose  $\beta_* > B$ . Lemma 3 supplies a principal factor holomorphic and bounded away from zero on  $\operatorname{Re} s > B$ . Lemma 2 and (4) give the direct bound required by the continuation criterion of [1, Section 2]. Its parameter  $\omega$  can be any sufficiently small positive number below  $\beta_* - B$ . Lemma 4 gives the other required bound with a target-independent saving. The criterion contradicts  $\beta_* > B$ . Hence  $\beta_* \leq B$ . The transfer in [1] from primitive Hecke characters to imprimitive Hecke and Dirichlet characters proves the stated assertions.  $\square$

## Verification scope

The rational inequalities in (4) and (7) can be checked independently by the accompanying `verify_bounds.py`. The file `Bounds.lean` checks their integer identities and two scaled linear bounds in Lean 4. It does not formalize the analytic lemmas, their extension to the new boundary, or the passage from real parameters to scaled integers. Compiling this note checks the TeX, not the number-theoretic inputs. The stronger boundary in Theorem 1 is not part of [1]’s stated theorem or its existing formalization.

## References

- [1] OpenAI, *The Quasi-Riemann Hypothesis: A Zero-Free Half-Plane*  $\operatorname{Re}(s) > 7/8$ , 30 September 2026. Source consulted 7 October 2026. [Source TeX](#).